

Master Theorem Worksheet

This is a worksheet to help you master solving recurrence relations using the Master Theorem. For each recurrence, either give the asymptotic solution using the Master Theorem (state which case), or else state that the Master Theorem doesn't apply. You should be able to go through these **25** recurrences in **10** minutes.

Problem 1-1. $T(n) = 3T(n/2) + n^2 f(n)$ $p = \log_2 3$ $n^2 > n^{\log_2 3}$
 $T(n) = \Theta(n^2) \leftarrow$ case 3

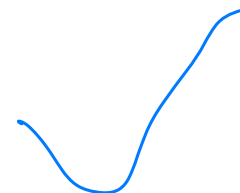
Problem 1-2. $T(n) = 7T(n/2) + n^2$
 $n^{\log_2 7} > n^2 \rightarrow \text{case 1}$
 $\Theta(n^{\log_2 7})$

Problem 1-3. $T(n) = 4T(n/2) + n^2$
 $n^{\log_2 4} = n^2 \rightarrow \text{case 2}$
 $\Theta(n^2 \log n)$

Problem 1-4. $T(n) = 3T(n/4) + n \lg n$
 $n^{\log_4 3} < n \lg n \rightarrow \text{case 3}$
 $\Theta(n \lg n)$

Problem 1-5. $T(n) = 4T(n/2) + \lg n$
 $n^{\log_2 4} > \lg n \rightarrow \text{case 1}$
 $\Theta(n^2)$

Problem 1-6. $T(n) = T(n-1) + n$
 doesn't apply



Problem 1-7. $T(n) = 4T(n/2) + n^2 \lg n$

$n^{\log_2 4} = n^2 < n^2 \log_2 n \rightarrow$ case 2
 $\theta(n^2)$ $\theta(n^2 \log^2 n)$

Handwritten notes: $n^{\log_2 4} \cdot \log^2 n \rightarrow$ case 2

Problem 1-8. $T(n) = 5T(n/2) + n^2 \lg n$

$n^{\log_2 5} > n^2 \log n \rightarrow$ case 1
 $\theta(n^{\log_2 5})$

Problem 1-9. $T(n) = 3T(n/3) + n/\lg n$

$n^{\log_3 3} = n > \frac{n}{\log n} \rightarrow$ case 1
 $\theta(n)$

Handwritten notes: not polynomially smaller & not case 2
 doesn't apply

Problem 1-10. $T(n) = 2T(n/4) + c$

$n^{\log_4 2} = n^{\frac{1}{2}} > c \rightarrow$ case 1
 $\theta(n^{\frac{1}{2}})$

Problem 1-11. $T(n) = T(n/4) + \lg n$

$n^{\log_4 1} = n^0 = 1$
 $\log n = n^0 \log n \checkmark \rightarrow$ case 2
 $\theta(\log^2 n)$

Problem 1-12. $T(n) = T(n/2) + T(n/4) + n^2$

doesn't apply

Problem 1-13. $T(n) = 2T(n/4) + \lg n$

$n^{\log_4 2} = n^{\frac{1}{2}} > \log n \rightarrow$ case 1
 $\theta(n^{\frac{1}{2}})$

Problem 1-14. $T(n) = 3T(n/3) + n \lg n$

$n^{\log_3 3} = n < n \log n = n^1 \log^1 n \rightarrow$ case 2
 $\theta(n \log^2 n)$

Handwritten notes: equal

Problem 1-15. $T(n) = 8T((n - \sqrt{n})/4) + n^2$

doesn't apply

Problem 1-16. $T(n) = 2T(n/4) + \sqrt{n}$

$$n^{\log_4 2} = n^{\frac{1}{2}} = \sqrt{n} \rightarrow \text{case 2}$$

$$\Theta(\sqrt{n} \log n)$$

Problem 1-17. $T(n) = 2T(n/4) + n^{0.51}$

$$n^{\frac{1}{2}} < n^{0.51} \rightarrow \text{case 3}$$

$$\Theta(n^{0.51})$$

Problem 1-18. $T(n) = 16T(n/4) + n!$

$$n^{\log_4 16} = n^2 < n! \rightarrow \text{case 3}$$

$$\Theta(n!)$$

Problem 1-19. $T(n) = 3T(n/2) + n$

$$n^{\log_2 3} > n \rightarrow \text{case 1}$$

$$\Theta(n^{\log_2 3})$$

Problem 1-20. $T(n) = 4T(n/2) + cn$

$$n^{\log_2 4} = n^2 > cn \rightarrow \text{case 1}$$

$$\Theta(n^2)$$

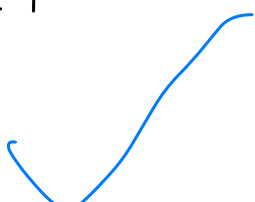
Problem 1-21. $T(n) = 3T(n/3) + n/2$

$$n^{\log_3 3} = n \simeq \frac{n}{2} \rightarrow \text{case 2}$$

$$\Theta(n \log n)$$

Problem 1-22. $T(n) = 4T(n/2) + n/\lg n$

$$n^{\log_2 4} = n^2 > \frac{n}{\lg n} \rightarrow \text{case 1}$$

$$\Theta(n^2)$$


Problem 1-23. $T(n) = 7T(n/3) + n^2$

$$n^{\log_3 7} < n^2 \rightarrow \text{case 3}$$

$$\Theta(n^2)$$

Problem 1-24. $T(n) = 8T(n/3) + 2^n$

$$n^{\log_3 8} < 2^n \rightarrow \text{case 3}$$

$$\Theta(2^n)$$

Problem 1-25. $T(n) = 16T(n/4) + n$

$$n^{\log_4 16} = n^2 > n \rightarrow \text{case 1}$$

$$\Theta(n^2)$$
